

Study of the Rifampin Monoresistance Mechanism in *Mycobacterium tuberculosis*

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Rifampin (RIF) susceptibility is a key factor in determining the treatment effectiveness of the standardized treatment regimens. In *Mycobacterium tuberculosis*, both target gene mutation and the efflux pump play major roles in the resistance to antituberculosis drugs. By eliminating RIF-resistant strains with *rpoB* mutation, the choice of RIF-monoresistant strains may allow us to identify the RIF-specific efflux pump genes. This study explored the RIF monoresistance mechanism in *M. tuberculosis*. Data from DNA sequencing and MIC measurements revealed that specific mutations, including Ser531Leu and His526Asp in RpoB, show high-level drug resistance. Three-dimensional structure modeling provided further evidence that the affinity between RIF and RpoB mutants was in accordance with the drug resistance level of the corresponding isolates. Furthermore, transcription-level analysis among the nonmutated isolates indicated that three efflux pumps (Rv0783, Rv2936, and Rv0933) might be involved in exporting RIF from the cell. Compared to 8 µg/ml for wild-type *Escherichia coli*, the MICs for the transgenic *E. coli* strains with either Rv0783 or Rv2936 were 32 and 16 µg/ml, respectively. In conclusion, our study indicated that several RpoB mutant types, including Ser531Leu and His526Asp, show high-level RIF resistance attributed to low affinity between RpoB mutant proteins and RIF. In addition, this work demonstrates that Rv2936 and Rv0783 may be responsible for low-level resistance to RIF by exporting RIF from cells. The predicted structure of RpoB and the newly identified efflux pumps in this study will provide a novel approach to design new drugs and develop novel diagnosis technologies.

Tuberculosis (TB), one of the most widespread and lethal infectious diseases worldwide, caused 8.8 million incident TB cases and 1.45 million deaths in 2010 (1). With the introduction of effective antimycobacterial drugs about half a century ago, the prevalence of TB appeared to be under control (2). However, the emergence of drug-resistant TB, especially multidrug-resistant (MDR) TB, defined as resistance to at least isoniazid and rifampin (RIF), has hampered effective TB treatment and control (3). A large number of recent studies have focused on the mechanisms of drug resistance. Several molecular and genetic mechanisms have been discovered to cause the resistance to most routine anti-TB drugs, including RIF (*rpoB*), isoniazid (*katG* and *inhA*), and streptomycin (*rspL* and *rrs*) (4–6).

RIF is the most important first-line antituberculosis drug and is a key factor in determining the treatment effectiveness of the treatment regimens (7, 8). Because more than 90% of RIF-resistant strains are also resistant to isoniazid, RIF resistance can be used as a valuable surrogate marker for MDR TB (9, 10). The mechanism of action of RIF is to arrest DNA-directed RNA synthesis of *Mycobacterium tuberculosis* by interacting with the β subunit of RNA polymerase (RNAP) (8, 11). Previous research has demonstrated that *rpoB* mutations of 95% of strains with RIF resistance are more likely located in the 81-bp region (codons 507 to 533) called the RIF resistance-determining region (RRDR) (12, 13). Inside the 81-bp RRDR, mutations within codons 516, 526, and 531 are responsible for up to 90% of RIF-resistant strains (14, 15). However, not all mutations within the RRDR display the same loss of RIF susceptibility (16, 17). The amino acid alterations of codon 526 or codon 531 cause high-level resistance to RIF, the MIC of which is greater than 32 µg/ml. In contrast, mutations in codons 511, 516, 518, and 522 cause low-level resistance to RIF (8, 17). Outside the RRDR, RIF-resistant mutation is also seen in the

amino-terminal region of *rpoB*, where the mutation in codon 176 results in high-level resistance to RIF (MIC of 1 to 32 µg/ml) (8).

In addition to mutations in the RRDR of the *rpoB* gene, an efflux pump may be responsible for approximately 5% of clinical RIF-resistant *M. tuberculosis* strains with no mutation in the RRDR (4). Efflux pumps, by which various molecules are exported outside the bacteria cell, are involved in the drug resistance described in several mycobacterial species (18), and the design of new therapeutic strategies may depend on the characterization of efflux pumps (19). Several putative efflux pumps have been reported to play a role in RIF resistance in *M. tuberculosis* based on large-scale transcriptional data (4, 20).

In the past few years, several studies have reported on the relationship between the RIF resistance level and the codon mutation *rpoB* (16, 17), while no explanation has been offered for the drug resistance difference mentioned above. Hence, it is meaningful to gain an insight into RIF resistance mechanisms, which will provide a novel track to design new drugs and develop new diagnosis technologies (4). By eliminating RIF-resistant strains with an *rpoB* mutation, the choice of RIF-monoresistant strains may allow us to identify the RIF-specific efflux pump genes. The aim of the present

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study was to explain the mechanism of RIF resistance in RIF-monoresistant strains for the first time, including both the classical mutations in known target genes and efflux pumps.

MATERIALS AND METHODS

Bacterial strains and culture conditions. Bacterial strains were all isolated from tuberculosis epidemiology surveillance in China. All bacterial cells were stored in Trypticase soy broth containing glycerol at -70°C . Prior to characterization of the RIF-resistant isolates, the strains were recovered on Lowenstein-Jensen medium for 4 weeks at 37°C .

Conventional drug susceptibility testing and *Mycobacterium* species identification. Four first-line anti-TB drugs (isoniazid, RIF, ethambutol, and streptomycin) and two second-line anti-TB drugs (kanamycin and ofloxacin) were used in studies of conventional *Mycobacterium* identification and testing of drug susceptibility toward clinical TB strains, which were performed as recommended by the World Health Organization (WHO) and the International Union Against Tuberculosis and Lung Disease (IUATLD) (1). The concentrations of drugs in medium were as follows: isoniazid, $0.2\text{ }\mu\text{g/ml}$; RIF, $40\text{ }\mu\text{g/ml}$; ethambutol, $2\text{ }\mu\text{g/ml}$; streptomycin, $4\text{ }\mu\text{g/ml}$; kanamycin, $30\text{ }\mu\text{g/ml}$; and ofloxacin, $2\text{ }\mu\text{g/ml}$. Medium supplied separately with paranitrobenzoic acid (500 mg/ml) and thiophen-2-carboxylic acid hydrazide (5 mg/ml) was used to perform *Mycobacterium* species identification. All drugs were purchased from Sigma-Aldrich (St. Louis, MO).

Genomic DNA extraction. Genomic DNA was extracted from freshly cultured bacteria. After being transferred into a microcentrifuge tube containing $500\text{ }\mu\text{l}$ Tris-EDTA (TE) buffer, the cells were centrifuged at $13,000\text{ rpm}$ for 2 min. The supernatant was discarded, and the pellet was resuspended in $500\text{ }\mu\text{l}$ TE buffer and then heated in a 95°C water bath for 1 h. After centrifugation of cellular debris, DNA in the supernatant was used for PCR amplification (21).

PCR amplification and sequencing of the *rpoB* gene. A 450-bp region of the *rpoB* gene containing the 81-bp RRDR was amplified by PCR. The forward primer was $5'\text{-ACCGACGACATCGACCACTT-3'}$, and the reverse primer was $5'\text{-GTACGGCGTTTCGATGAACC-3'}$. The content of $50\text{ }\mu\text{l}$ of PCR mixture was $5\text{ }\mu\text{l}$ of PCR buffer, 2 mM MgCl_2 , $200\text{ }\mu\text{M}$ each deoxynucleoside triphosphate (dNTP), $0.2\text{ }\mu\text{M}$ each primer, $2\text{ }\mu\text{l}$ of genomic DNA, and $0.5\text{ }\mu\text{l}$ of AmpliTaq DNA polymerase. The PCR programs were performed with an initial denaturation at 95°C for 5 min, followed by 35 cycles of denaturation at 94°C for 1 min, annealing at 58°C for 1 min, extension at 72°C for 30 s, and final extension at 72°C for 7 min. The primers were synthesized by Invitrogen. After purification with a QIAquick gel extraction kit, the amplicons were sequenced with an Applied Biosystems ABI Prism BigDye terminator cycle sequencing kit with an ABI Prism 3130 genetic analyzer.

Determination of MIC. To determine the MICs of RIF-monoresistant *M. tuberculosis* strains identified by conventional DST, a microplate alamarBlue assay (MABA) was performed as described previously (22). Final RIF concentrations were 0.125 to $512\text{ }\mu\text{g/ml}$. The MICs were defined as the lowest concentration of antibiotic that reduced the viability of the culture by at least 90%, as determined by fluorescence measurements at room temperature in top-reading mode, in which the excitation wavelength and emission wavelength were 530 nm and 590 nm , respectively. The MIC breakpoint concentration for RIF was defined as $0.5\text{ }\mu\text{g/ml}$. The protocol for detection of the MIC of *Escherichia coli* was referred to in the previous study (19).

Molecular typing methods. The spoligotyping method was performed using primers DRa and DRb as previously reported (23). ECL enhanced chemiluminescence detection liquid was used to detect hybridizing DNA, followed by exposure to X-ray film. The original spoligotyping patterns were converted into octal formats. The mycobacterial interspersed repetitive unit-variable number of tandem repeat (MIRU-VNTR) typing method was performed with the primers in accordance with the standardized protocol (24). The PCRs for all MIRU-VNTR loci were performed in a reaction volume of $20\text{ }\mu\text{l}$ containing $10\text{ }\mu\text{l}$ of $2\times$ Taq

mixture, $2\text{ }\mu\text{l}$ of DNA template, and 5 pmol of each primer set. The PCR amplification program was 94°C for 3 min, followed by 32 cycles at 94°C for 1 min, 60°C for 1 min, and 72°C for 1 min, with a final extension at 72°C for 5 min. Next, $3\text{ }\mu\text{l}$ of amplicon from the reaction tube was run on a 1.5% agarose gel, and a 100-bp ladder was run every six lanes. The images were collected and analyzed using Quantity One software (Bio-Rad).

Modeling of three-dimensional structure for RpoB. Modeling of protein structures and analysis of protein-ligand interaction were carried out by Discovery Studio 3.1 (Accelrys). The wild-type protein structure with an RIF ligand was modeled from residues 112 to 1226, using a Build Homology Models module with Protein Data Bank (PDB) (<http://www.rcsb.org/pdb/>) file 1I6V (13) as the template. After refinement by the Loop Refinement (MODELER) module, a Smart Minimizer algorithm for energy minimization was performed to an RMS gradient of $0.0001\text{ kcal}/(\text{mol} \times \text{\AA})$ with the protein backbone atoms fixed. The Build Mutants module was utilized to produce the structure models of Ser531Leu, His526Asp, His526Gly, His526Leu, His526Arg, and Leu533Pro; the Align and Superimpose Proteins module was used to compare the mutant structures with the wild-type structure. The interactions between RIF and wild-type or mutant RpoB were analyzed using the Analyze Ligand Interactions and Structure Monitor modules.

qRT-PCR. Bacteria were harvested from 7H9 medium with oleic acid-albumin-dextrose-catalase (OADC) supplement after growing for 4 weeks. RNA was isolated by TRIzol according to the instructions of the manufacturers. After treatment with DNase I (Invitrogen), the cDNAs were reverse transcribed from $5\text{ }\mu\text{g}$ of total RNA with a SuperScriptIII reverse transcription (RT) kit (Invitrogen). The quantitative reverse transcription-PCR (qRT-PCR) was performed in a $20\text{-}\mu\text{l}$ system containing $10\text{ }\mu\text{l}$ of $2\times$ the mixture supplied with SYBR green (New England Laboratories), $2\text{ }\mu\text{l}$ of cDNA template, and 5 pmol of each primer set. The primers for qRT-PCR analysis are listed in Table 1. Poly(A) was used as the internal control in respective PCR experiments.

Data analysis. The genotyping data were analyzed using BioNumerics 5.0 (Applied Maths, Belgium) software. Cluster analysis was performed, and a dendrogram was generated in BioNumerics using the similarity coefficient and unweighted-pair group method using average linkages (UPGMA) coefficient. In addition, SPSS 14.0 (SPSS, Inc.) was used to perform chi-square analysis, and differences were considered to be statistically significant when $P < 0.05$.

RESULTS

Identification of *rpoB* mutation. RIF-monoresistant clinical isolates (24 strains) from the national baseline surveillance were examined for RIF MIC and *rpoB* sequence. Among the 24 strains monoresistant to RIF, the MICs of 13 isolates were $>32\text{ }\mu\text{g/ml}$, while the MICs of the others were $<16\text{ }\mu\text{g/ml}$. Meanwhile, sequence details of the *rpoB* locus were obtained from the 24 mutant strains. Surprisingly, no mutation was found in the RRDR of the *rpoB* gene in more than 30% (8/24) of isolates. The *rpoB* mutation in the RRDR was observed in the other 16 isolates, including His526→Gly (CAC→GGC) (two isolates), His526→Asp (CAC→GAC) (two isolates), His526→Leu (CAC→CTC) (one isolate), His526→Arg (CAC→CGC) (one isolate), Ser531→Leu (TCG→TTG) (nine isolates), and Leu533→Pro (CTG→CCG) (one isolate) (Table 2). Codon 531 had the highest mutational frequency (9/24 [37.5%]), followed by codon 526 (6/24 [25.0%]) and codon 533 (1/24 [4.2%]) (Table 2).

Determination of the MIC. To better define the relationship between the different *rpoB* mutations and the RIF resistance, we analyzed the MIC according to different point mutations in the *rpoB* gene. The RIF MICs of strains with mutations at codons 531 and 526 ranged from $64\text{ }\mu\text{g/ml}$ to $256\text{ }\mu\text{g/ml}$ and $0.5\text{ }\mu\text{g/ml}$ to

TABLE 1 Primers used in this study for qRT-PCR amplification

Locus tag or gene product	Length (bp)	Primer
Rv2936	146	5'-TAGACATCGCGTGCGGATTGGT-3' F 5'-GCGTGGTCAACAACGTGGCAAT-3' R
Rv2937	232	5'-TCGCCAGCAACTTAGGGCAATACA-3' F 5'-TCCGATGACGTAGCCGCAACTAG-3' R
Rv2938	171	5'-GTTTGGTGCCGCTCAACTCGTATC-3' F 5'-GGTACGGCGCATACGACGCAGATA-3' R
Rv2846c	225	5'-CGCCCTACGGGAAACCAACAAAGA-3' F 5'-GCGGAACAAGTGGAAACGGCAGCAGAC-3' R
Rv3065	166	5'-TAGTGGGTTATGGCATCGCTTTTCG-3' F 5'-GACGCCAACCACCTTCATCACAGA-3' R
Rv0783c	147	5'-ACCGCACAGAACATCCGCTCATAG-3' F 5'-GATTGGTGCACACTTGCTGGAGG-3' R
Rv0849	93	5'-GTCGTTTCGCAACCGTCCGTTTCTG-3' F 5'-CCTGCATGGGCAGAGCCAGATAGA-3' R
Rv1145	241	5'-GACGACCTGCTGGTGATGGAGTTG-3' F 5'-CGACTGACGATGAGCAGCGTGATAG-3' R
Rv1146	181	5'-ATGTTCGCGCTCGGCCTGACTTTA-3' F 5'-GAACGTCTCCTCGAAACCGGCTCT-3' R
Rv1250	132	5'-GCAGCCTTGGATTGGGCGGTGAT-3' F 5'-GGACAAGCTGAAGTTCCGGTGCCTT-3' R
Rv1258c	117	5'-CGTCTGGAAACCTGCGGGTATTGCG-3' F 5'-CGGTTGCTGGTGGTTCGGTGAAGTA-3' R
Rv1410c	204	5'-ATCCCGACGGCAAACACGTACTGC-3' F 5'-ACATCAACCAGCGTCACCATCAGC-3' R
Rv1634	156	5'-TCGATACCTACGTGCCGCTGTTTCG-3' F 5'-GCTGCCACGACATGCCCCGATAACT-3' R
Rv1819c	396	5'-CGGCGACTTCCTTCGGCAACAT-3' F 5'-ACATACGGCAACTGCGACAAGAA-3' R
Rv2209	220	5'-CTGGGCACCACGTTCTTCAGC-3' F 5'-GCGTGAACCCACTGCCACA-3' R
Rv2994	146	5'-ATGCGTCCCGTCCGCCTGAT-3' F 5'-GGTGGCTTCTAGCCCGTTGTCC-3' R
Rv1877	234	5'-TGTGCTGCGTCTGCTTCCTTCGT-3' F 5'-AGGAACGCAACCGCCATCAG-3' R
Rv2333c	216	5'-TCCGATGATGGATCTGACCCTG-3' F 5'-GCCAACCAGGTGCCCAACA-3' R
Rv2459	212	5'-CGTCGCCCTGATCGCATACA-3' F 5'-CAGGACATCACCACGAAGTAGACG-3' R
Rv0933	94	5'-CTGACCCGACTACCACCGAGAA-3' F 5'-GCCTGGGCAAGGTTATGGGTC-3' R
Poly(A)	180	5'-GTCGTGGTTGGACCTTGGAGGG-3' F 5'-GCGTCCGTATCGTCGTCATCG-3' R

256 µg/ml, respectively. The RIF MICs of resistant strains without mutation ranged from 0.5 µg/ml to 128 µg/ml. For the resistant strains with the same mutant type, the MICs were similar, and the specific alterations, instead of all mutations, in codon 531 (Ser→Leu) and codon 526 (His→Asp) resulted in high-level resistance to RIF (MIC of >32 µg/ml) (Table 2). In contrast, most of the isolates with no mutation resulted in low-level resistance to RIF, except for two isolates, the MIC of which was 128 µg/ml (Table 2). As shown in Table 3, the average MIC of six mutants with Ser531→Leu was 192 µg/ml, followed by 256 µg/ml in two mutants with His526→Asp, 0.75 µg/ml in two mutants with His526→Gly, 4 µg/ml in one mutant with His526→Leu, 2 µg/ml in one mutant with His526→Arg, and 0.5 µg/ml in one mutant with Leu533→Pro. In addition, the average MIC with no mutation was 42.2 µg/ml.

TABLE 2 Mutation and MIC detected in RIF-monoresistant *M. tuberculosis* study isolates

Strain ID	Codon	Prevalence (%)	Mutation	MIC (µg/ml)
TB1	531	37.5	TCG(Ser)→TTG(Leu)	256
TB2			TCG(Ser)→TTG(Leu)	64
TB3			TCG(Ser)→TTG(Leu)	256
TB4			TCG(Ser)→TTG(Leu)	256
TB5			TCG(Ser)→TTG(Leu)	256
TB6			TCG(Ser)→TTG(Leu)	128
TB7			TCG(Ser)→TTG(Leu)	128
TB8			TCG(Ser)→TTG(Leu)	128
TB9			TCG(Ser)→TTG(Leu)	256
TB10	526	25.0	CAC(His)→GGC(Gly)	1
TB11			CAC(His)→GGC(Gly)	0.5
TB12			CAC(His)→GAC(Asp)	256
TB13			CAC(His)→GAC(Asp)	256
TB14			CAC(His)→CTC(Leu)	4
TB15			CAC(His)→CGC(Arg)	2
TB16	533	4.2	CTG(Leu)→CCG(Pro)	0.5
TB17	— ^a	33.3	—	0.5
TB18			—	0.5
TB19			—	0.5
TB20			—	8
TB21			—	8
TB22			—	64
TB23			—	128
TB24			—	128

^a —, no mutation.

Characterization of genotyping. The dendrogram (Fig. 1) based on MIRU-VNTR profiles illustrates the genetic relationship and correlations between genotype and resistance phenotype. As shown in Fig. 1, the minimum spanning tree (MST) of the RIF-monoresistant strains involved three major branches. The largest essentially contained the Ser531Leu mutation type and other strains with high-level RIF resistance, except for strain TB3. The other two branches predominantly included strains with low-level RIF resistance, including H37Rv. Interestingly, the genotype of drug-resistant strains showed similar profiles to the level of RIF resistance. Additionally, spoligotyping analysis revealed that 92.9% (13/14) of strains with high-level RIF resistance belonged to the Beijing family, while only 50% (5/10) of ones with low-level RIF resistance were of the Beijing genotype. The Beijing family was therefore more likely to be related to the strains with high-level RIF resistance than the non-Beijing family ($P = 0.017$).

TABLE 3 Effect of different *rpoB* mutations conferring RIF resistance as measured by MIC

<i>rpoB</i> mutation type	No. of mutants	MIC (µg/ml)
Ser531Leu	9	192
His526Asp	2	256
His526Gly	2	0.75
His526Leu	1	4
His526Arg	1	2
Leu533Pro	1	0.5

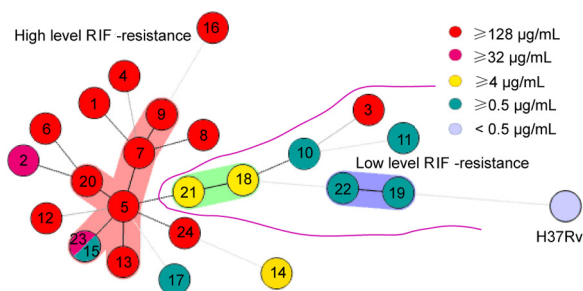


FIG 1 Dendrogram of 24 RIF-monoresistant *M. tuberculosis* isolates. The phylogenetic tree was generated with a 24-VNTR locus set. The colors represent different MICs, and the numbers indicate the strain identifier (ID).

Structure modeling and comparison of wild-type and mutant RpoBs. For further structural analysis, wild-type TB RpoB was modeled by Discovery Studio 3.1 using the crystal structure of *Thermus aquaticus* (*Taq*) core RNAP complexed with RIF as a template (13). The predicted structure of TB RpoB was very similar to the crystal structure of *Taq* core RNAP, and the root mean square deviation (RMSD) value of the main chain atoms between these two structures was 3.572 Å (Fig. 2A). As shown in Fig. 2B, the RRDR of TB RpoB (Gly507-Leu533) formed a sheet-helix-sheet motif, which made up the main structure of the RIF binding pocket. The RMSD value of the main chain atoms between the RRDR of TB *rpoB* and the corresponding region of *Taq* core RNAP was 0.505 Å. Twelve residues participated in the intermo-

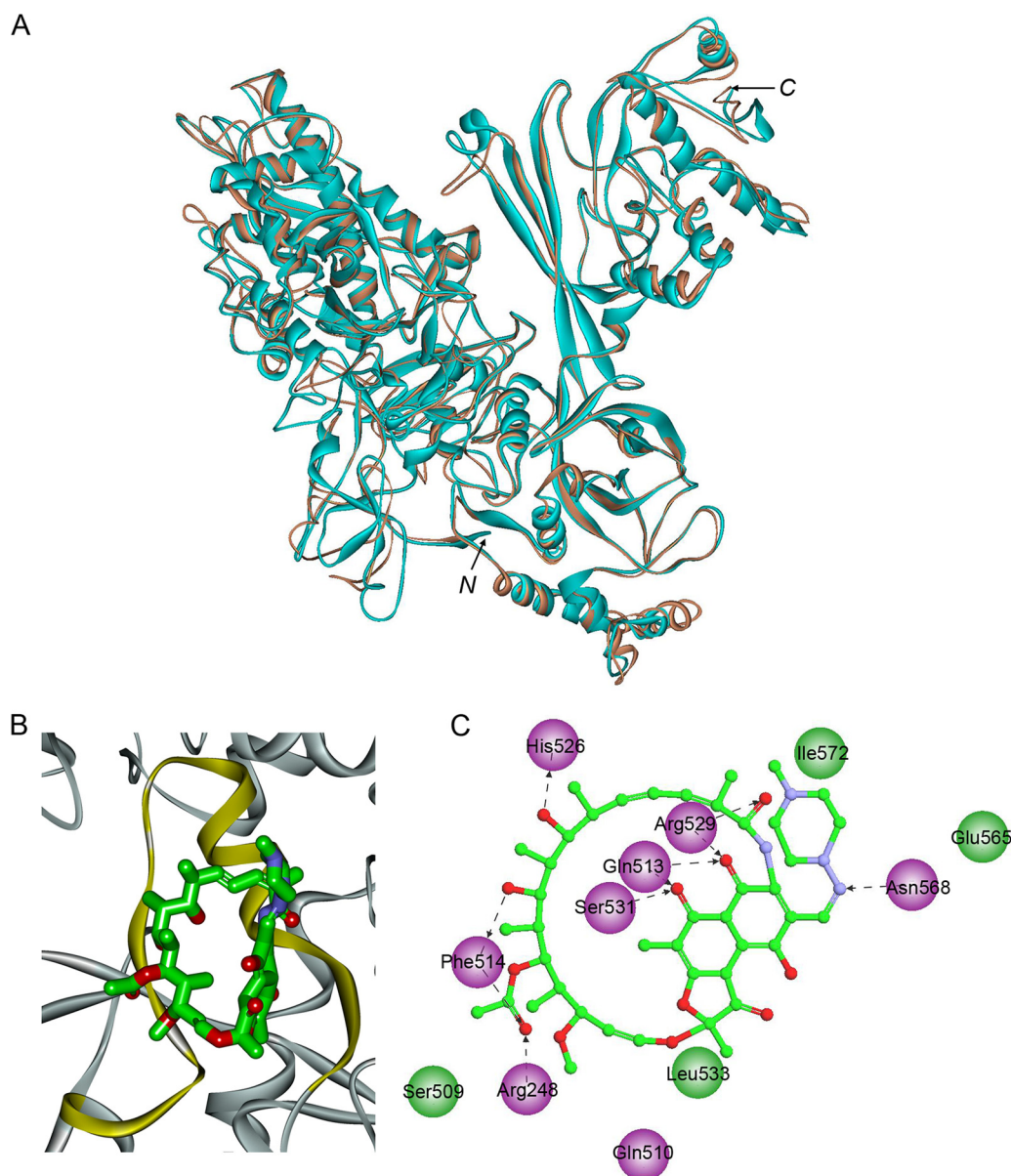


FIG 2 Structure model of wild-type TB RpoB and detailed interactions of the RIF-RpoB complex. (A) Superimposition of wild-type TB RpoB (light brown) and *Taq* core RNAP (cyan; PDB code 1I6V). (B) RIF (carbon atoms in green) binding pocket of wild-type RpoB, with the RRDR segment highlighted in yellow. (C) Detailed interactions between wild-type RpoB and RIF. The intermolecular hydrogen bonds are indicated by black dashed arrows. Residues involved in hydrogen bond, charge, or polar interactions are represented by magenta-colored balls, and residues involved in van der Waals interactions are represented by green balls.

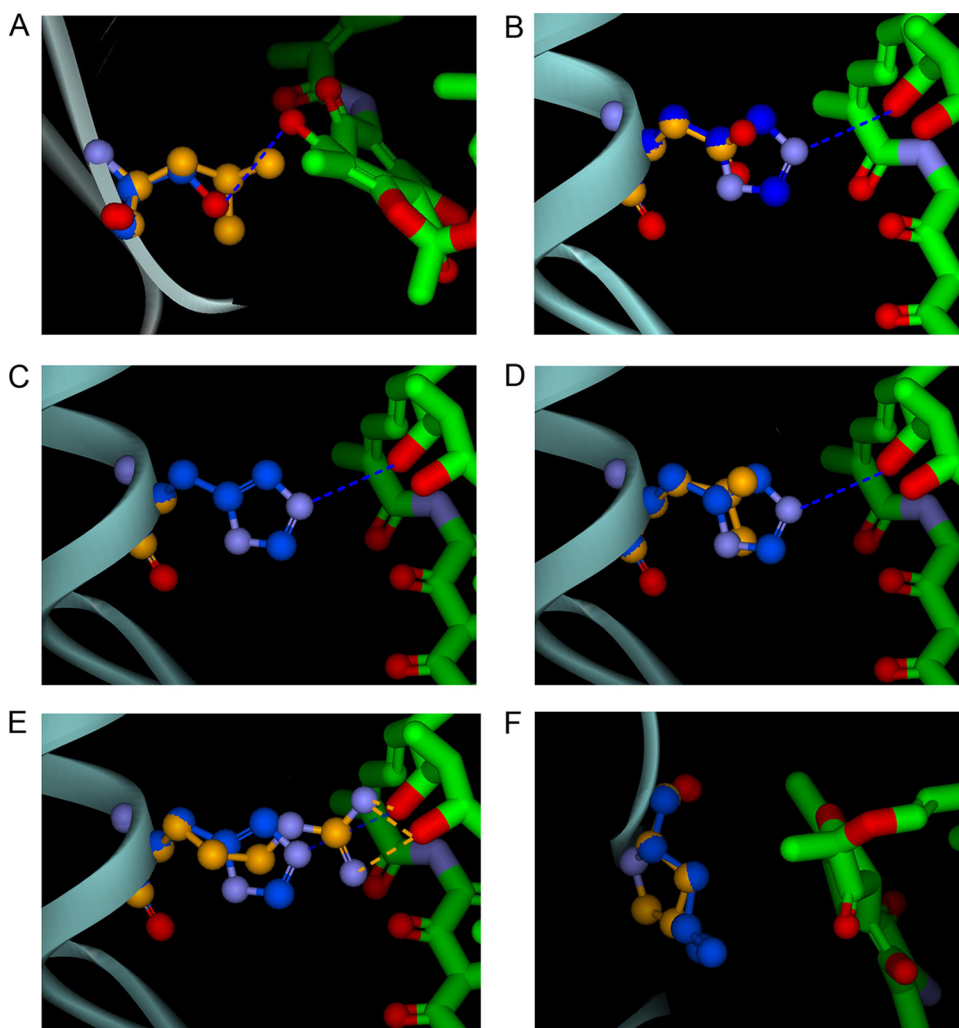


FIG 3 Structural comparisons between wild-type RpoB and the Ser531Leu (A), His526Asp (B), His526Gly (C), His526Leu (D), His526Arg (E), and Leu533Pro (F) mutants. The carbon atoms of wild-type and mutant RpoB are shown in blue and orange, respectively. The intermolecular hydrogen bonds in wild-type and mutant complexes are indicated by blue and orange dashed lines, respectively.

lecular interactions between RpoB and RIF (Fig. 2C), and eight of them were located in the RRDR, including the His526, Ser531, and Leu533 residues analyzed in this study.

The structural comparisons between wild-type RpoB and mutant RpoBs resistant to RIF are shown in Fig. 3. In wild-type RpoB, there is an intermolecular hydrogen bond between the side chain of Ser531 and the RIF molecule. In the Ser531Leu mutant, the replacement of Ser531 by Leu531 disrupted the above-mentioned hydrogen bond and led to structural bumps between the side chain of Leu531 and the RIF molecule (Fig. 3A). His526 in wild-type RpoB had a positive charge opposite the nearby RIF surface, and its side chain formed an intermolecular hydrogen bond with the RIF molecule. After replacement by Asp526, the charge character of the residue changed from positive to negative, which could repel the negative charge on the RIF surface, and the hydrogen bond in the wild-type RpoB-RIF complex was disrupted (Fig. 3B). The replacement of His526 by a neutral residue of similar (Leu) or smaller (Gly) size only disrupted the hydrogen bond between wild-type His526 and the RIF molecule and didn't lead to a structural bump (Fig. 3C and D). In the His526Arg mutant, the

526Arg residue had hydrogen bond contact with and electrostatic attraction to the RIF molecule, but the larger size of the Arg residue led to structural bumps (Fig. 3E). In the wild-type RpoB-RIF complex, Leu533 contacted the RIF molecule through van der Waals interactions. The replacement of Leu533 by Pro neither changed the interaction nor introduced structural bumps (Fig. 3F). The changes in intermolecular interactions between mutant residues and the RIF molecule from the wild type to the mutants are summarized in Table 4.

Identification of efflux pump. We selected the RIF-monoresistant isolates without mutations in the RRDR to investigate the transcriptional level of 20 efflux pump genes related to drug resistance. Three efflux pump genes (including Rv2936, Rv0783, and Rv0933) were found to be overexpressed among clinical isolates with high-level resistance, in accordance with the level of drug resistance (Fig. 4). The other 17 pumps revealed no significant differences among various drug-resistant strains. To further assess the function of the three candidate efflux pump genes conferring RIF resistance, real-time PCR experiments were performed in seven MDR strains, and the data revealed that there were no dif-

TABLE 4 Changes in intermolecular interactions between mutant residues and RIF molecules from the wild type to mutants^a

Mutant	Hydrogen bond	van der Waals interaction	Electrostatic repulsion	Structural bumps
Ser531Leu	Yes→no	No→yes		No→yes
His526Asp	Yes→no		No→yes	
His526Gly	Yes→no			
His526Leu	Yes→no			
His526Arg				No→yes
Leu533Pro				

^a Changes from wild type to mutant are represented from left to right, respectively.

ferences in the transcriptional level (see Fig. S1 in the supplemental material).

The *E. coli* strain transformed with expression vector pEASY-E1 as a control showed an initial drug resistance to RIF, the MIC of which was 8 µg/ml. The MICs of *E. coli* clones transformed with pEASY-E1-Rv2936 and pEASY-E1-Rv0783 were 32 µg/ml and 16 µg/ml, respectively. In comparison, transformation of the *E. coli* clones with pEASY-E1-Rv0933 revealed no effect on RIF resistance, and the MIC was still 8 µg/ml (Table 5).

DISCUSSION

Resistance to RIF is one of the major reasons for treatment failure and fatal clinical outcome in TB patients (7, 8). Both classical mutations in known target genes and efflux pumps confer drug resistance in *M. tuberculosis* strains. As previously reported, mutations in the RRDR are responsible for resistance in up to 95% of RIF-resistant strains. In this study, RRDR mutations were detected in two-thirds of clinical isolates (66.7%; *n* = 16), while more than 30% of strains harbored no mutation in the RRDR. One possible explanation for this is that only RIF-monoresistant isolates, instead of all RIF-resistant isolates, were collected in this study. In addition, the most frequent mutations in RIF-monore-

TABLE 5 *In vitro* RIF susceptibility of overexpression of Rv2936, Rv0783, and Rv0933 genes as measured by MIC

Construct in <i>E. coli</i>	MIC (µg/ml)
pET28a	8
pET28a-Rv2936	32
pET28a-Rv0783	16
pET28a-Rv0933	8

sistant strains were located in codons 526 and 531, and Ser531Leu was the most dominant mutation type, in accordance with previous reports (25, 26).

In the present study, the dendrogram analysis revealed that most strains with low-level resistance were clustered in one clade, while strains with high-level resistance were revealed in other clades. Also, strains with low-level resistance appeared to be less polymorphic than ones with high-level resistance. In this study, a drug-resistant strain was always the product of an anti-TB drug, and selection at low concentrations of the drug produced many mutants with low-level resistance. As selection pressure increased, a mixture of variants became prevalent, which has been proven by fluoroquinolone supplementation in *M. tuberculosis* (27). Hence, a high drug concentration might be responsible for high diversity of *M. tuberculosis* strains. As strains with high-level resistances are generated from high drug concentration pressure, they will show more polymorphism than ones with low-level resistance. Interruption of anti-TB treatment is a strong predictor of drug-resistant TB, especially MDR (28). Following several months of selection of high concentrations of antibiotics in treatment-interrupted patients, the TB population may show high diversity, including drug-resistant colonies. In contrast, completion of the course of anti-TB treatment would kill variants effectively, instead of producing “super extensively drug-resistant tuberculosis.”

The binding affinity between a ligand and its receptor protein is

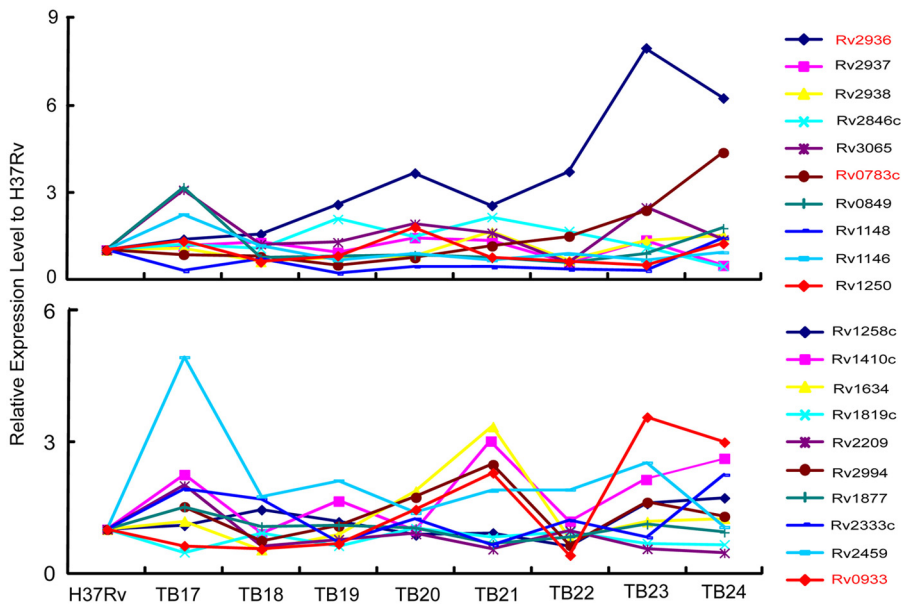


FIG 4 Relative expression levels of 20 different putative efflux pump genes in 8 *M. tuberculosis* isolates. qRT-PCR was performed in triplicate using independent RNA samples prepared from different strains. The points on the line chart indicate fold increase relative to the H37Rv value (arbitrarily set to 1) of the same efflux pump gene.

mainly dependent on the structural complement and intermolecular interaction. Both a good structural complement without any structural bumps and a strong intermolecular interaction, such as hydrogen bonds and electrostatic interaction, led to high binding affinity between the ligand and receptor. Both the mutations Ser531→Leu and His526→Asp disrupted the intermolecular hydrogen bonds between mutation residues and the RIF molecule. In addition, the replacement of Ser531 by Leu531 led to structural bumps and the replacement of His526 by Asp526 introduced charge repulsion, which further reduced the affinity between RIF and RpoB. Thus, Ser531Leu and His526Asp mutants have low affinities to the RIF molecule, and the strains with these mutant RpoBs showed high-level RIF resistance. Although the mutation His526→Arg led to a structural bump between the mutated residue and RIF, the Arg residue has the same charge as the His residue, which maintained the hydrogen bond interaction and charge affinity between the mutated residue and RIF. The mutations His526→Leu and His526→Gly only disrupted the hydrogen bond interaction, and the mutation Leu533→Pro did not significantly change the intermolecular interaction between RpoB and RIF. Therefore, compared to wild-type RpoB, mutations His526→Gly, His526→Leu, His526→Arg, and Leu533→Pro did not significantly reduce the affinity between the RpoB and RIF, and the strains with these mutations showed low-level RIF resistance.

In addition to *rpoB* mutation, the involvement of an active efflux pump mechanism is also responsible for RIF resistance in *M. tuberculosis* (20). Several efflux pumps have been supposed to play a role in low-level RIF resistance, including Rv1258c, Rv1410c, and Rv0783, which belong to the major facilitator superfamily (MFS) (29–33). In accordance with previous literature, our data revealed that the increased transcription of Rv0783 might confer resistance to RIF. We also discovered that Rv2936 might serve as an RIF-specific efflux pump. In the present study, the transcriptional level of Rv0933 was significantly upregulated in the RIF-monoresistant strains, while no increase of RIF MIC was observed when overexpressing Rv0933 in *E. coli*. The Rv0933 gene, encoding an ABC transporter, is hypothesized to confer phosphate transportation, which is essential for the survival of *M. tuberculosis* (12, 34). Hence, one possible explanation for highly expressed levels of Rv0933 in drug-resistant strains is that Rv0933 may be involved in RIF resistance by improving the nutritional status of bacteria, while the auxiliary role of Rv0933 in the drug resistance only emerges when other efflux pumps related to drug resistance exist.

However, there are several questions that need to be resolved. First, the sample size of RIF-monoresistant isolates in this study is small, and several mutant types are not included. Although the binding patterns of RIF to mutant proteins can be predicted, their drug-resistant phenotypes need to be declared. In addition, the transgenic experiments were carried out with *E. coli* instead of *Mycobacterium smegmatis*, a species related to *M. tuberculosis*. According to previous literature (28), *E. coli* has been introduced to determine the exogenous gene's function conferring RIF resistance in *M. tuberculosis*. Hence, it may be also reasonable to perform this technique in this study. However, further studies on the specific efflux pumps need to be performed with *M. smegmatis*. While there are some shortcomings in the present study, resolving them will further help us to reveal the RIF resistance mechanism in *M. tuberculosis*.

In conclusion, we have successfully performed a comprehensive study of the RIF resistance mechanism of *M. tuberculosis*. Specific mutant types located in the RRDR, including the Ser531Leu and His526Asp mutants, show high-level drug resistance due to low affinity between RpoB mutant proteins and RIF. The predicted three-dimensional (3D) structures of RpoB mutants described here provide us the molecular rationale for strategies to modify RIF, which will inhibit the growth of the RIF-resistant strains more efficiently. In addition, this work demonstrates that Rv2936 and Rv0783 may serve as RIF-related efflux pumps, conferring low-level resistance to RIF. Further study will be carried out to identify their function as RIF-resistant efflux pumps among a large number of *M. tuberculosis* isolates.

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